Team Name : Light Curve AI

Team Leader Name : Divyansh Mehta

Problem Statement : AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves

Team Members

Team Leader:

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Team Member-1:

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Team Member-2:

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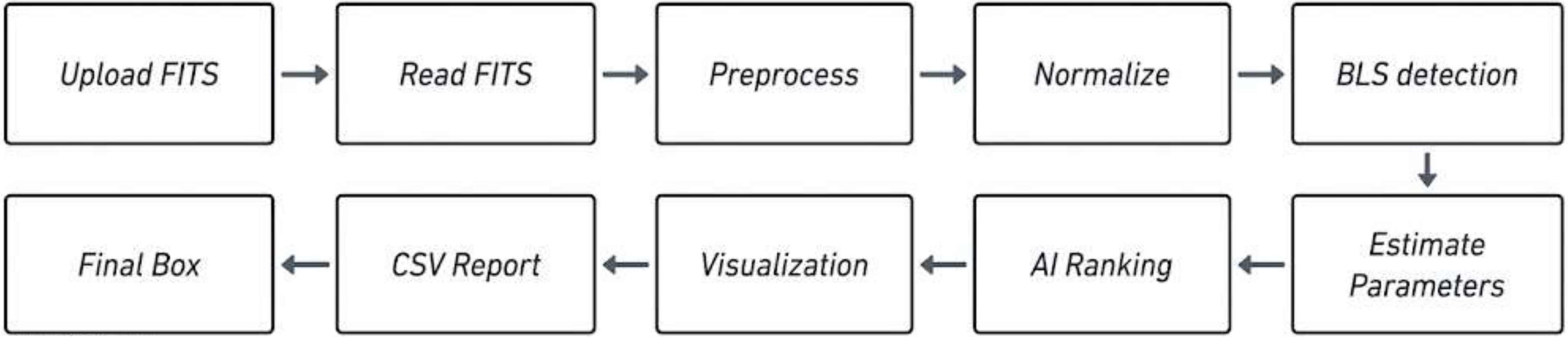
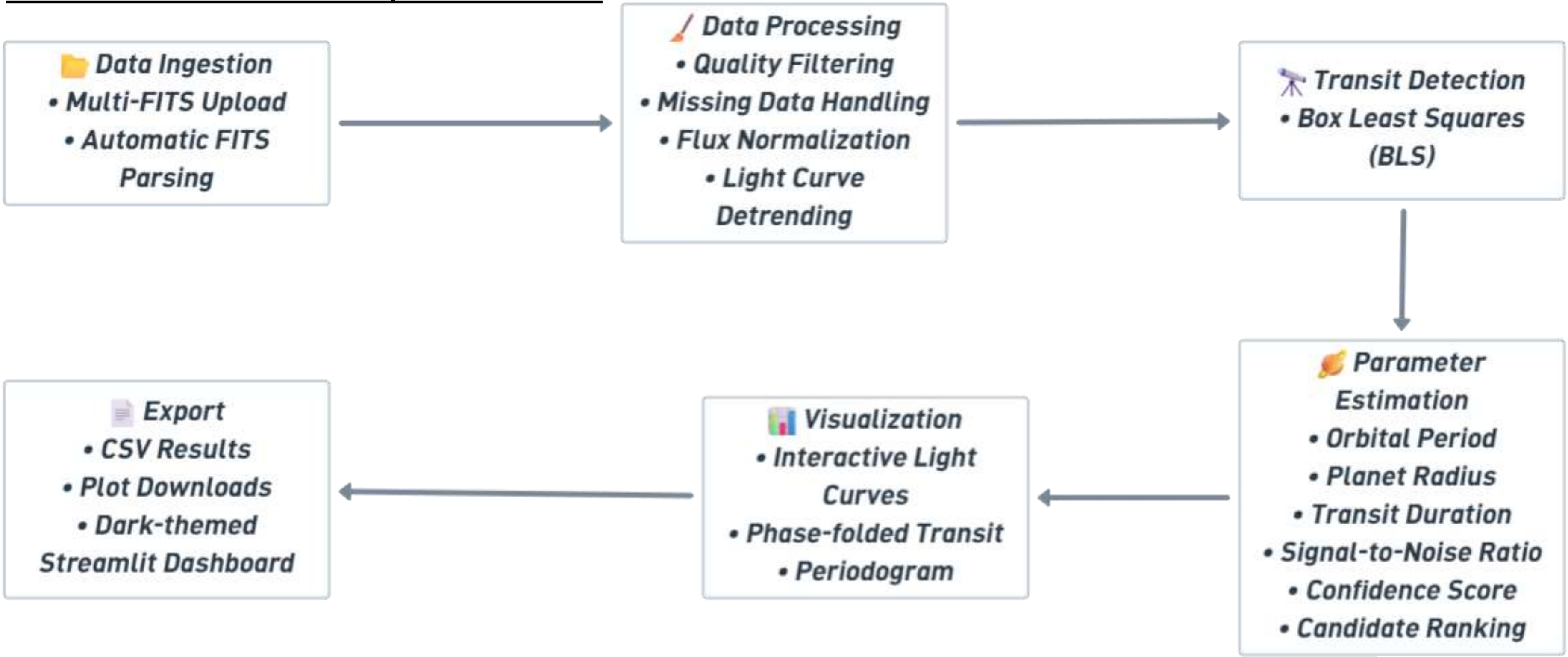
Opportunity should be able to explain the following:

- Our proposed solution combines classical astronomical signal processing with artificial intelligence in a single automated pipeline
- The solution automates the complete exoplanet detection workflow, significantly reducing the manual effort required to inspect thousands of stellar light curves.
- AI-assisted candidate classification and confidence scoring
- Automated generation of plots and CSV reports
- Open-source architecture using widely adopted astronomy libraries

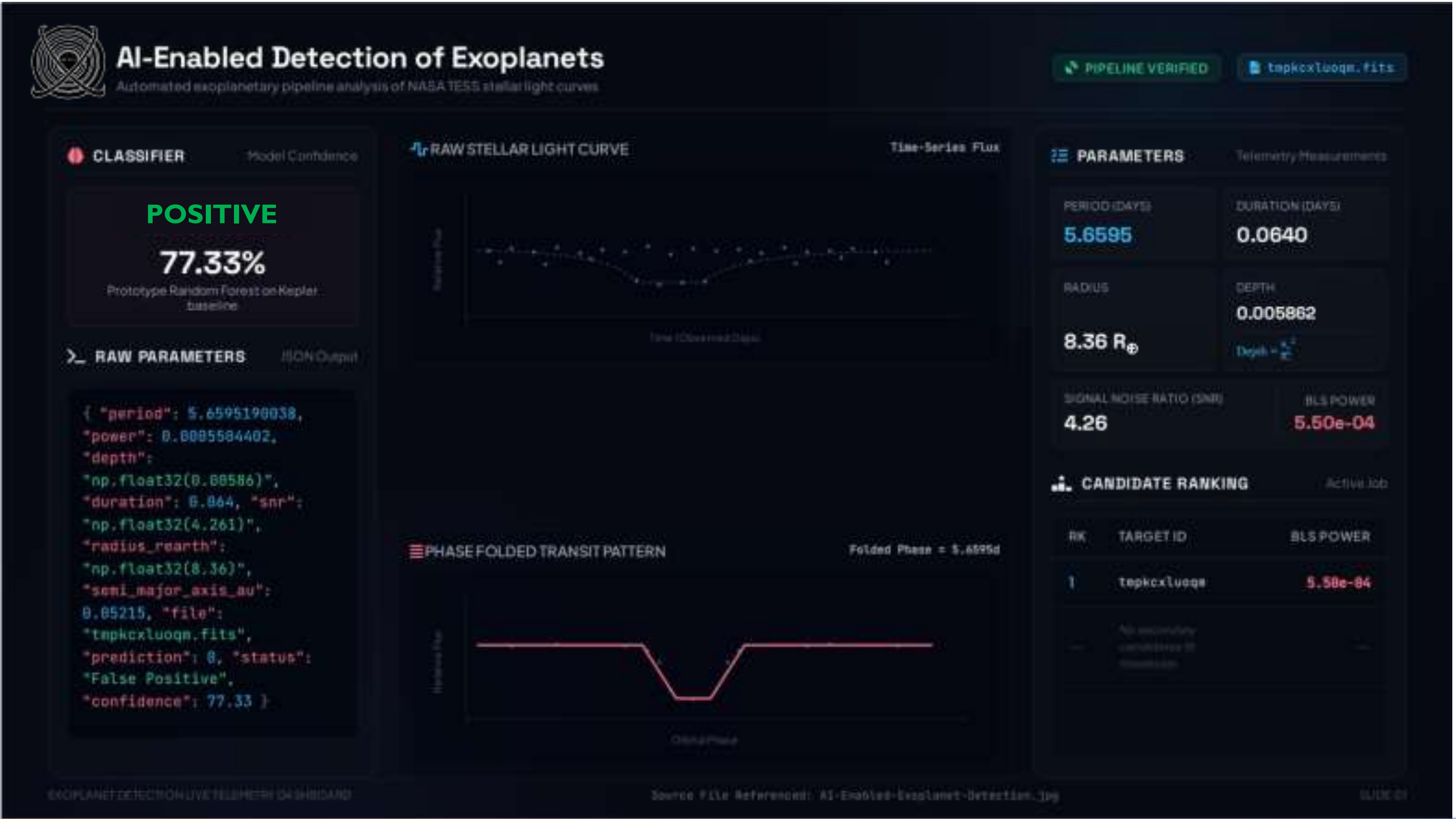
Process flow diagram:



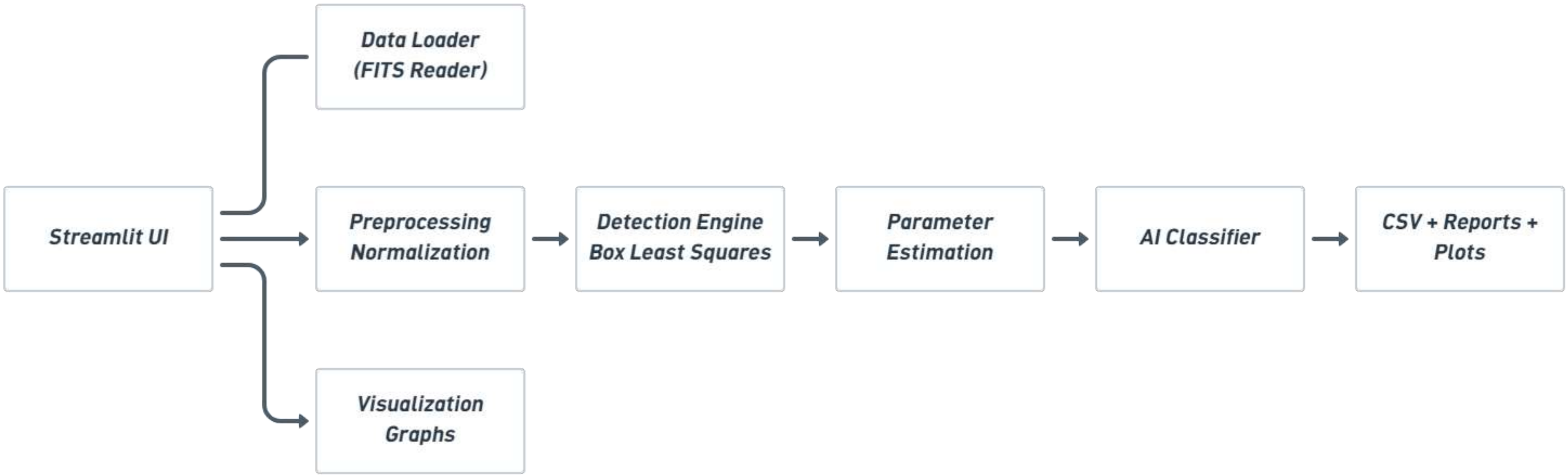
List of features offered by the solution:



Wireframe of the proposed solution:



Architecture diagram of the proposed solution



Technologies to be used in the solution:

Python

NumPy

Matplotlib

Streamlit

Pandas

Plotly

Astropy

SciPy

NASA TESS

Lightkurve

Scikit-learn

CNN

Estimated implementation cost (optional):

Component	*Estimated Cost (₹)*
High-Performance Workstation	*1,20,000*
Cloud Computing & GPU Resources	*45,000*
Data Storage & Backup Infrastructure	*15,000*
Software Development & Integration	*40,000*
UI/UX Development & Dashboard Design	*20,000*
Testing, Validation & Quality Assurance	*18,000*
Documentation & Technical Report Preparation	*12,000*
Deployment, Maintenance & Miscellaneous	*30,000*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